

REM2 Research Methods

Exercise 5-1: Standard diagnostic test in HPM

Instructions: Work in groups of two to complete the following tasks

1. Open “Lectures 4-6 Data.xlsx” in a software of your choice. Estimate the following regression model by using the rental market data only.

$$\begin{aligned} \ln(\text{Price}) = & \beta_0 + \beta_1 \ln(\text{Floorarea}) + \beta_2 \ln(\text{Busstation}) + \beta_3 \ln(\text{Railstation}) \\ & + \beta_4 \text{Airport} + \beta_5 \text{Bedrooms} + \beta_6 \text{Bathrooms} \\ & + \beta_7 \text{Terraced House} + \beta_8 \text{Semi - Detached House} \\ & + \beta_9 \text{Bungalow} + \beta_{10} \text{Detached House} + \beta_{11} \text{Year}_{2018} \\ & + \beta_{12} \text{Year}_{2019} + \beta_{13} \text{Year}_{2020} + \beta_{14} \text{Year}_{2021} + \beta_{15} \text{Year}_{2022} + \varepsilon \end{aligned}$$

2. Generate the predicted values and residuals from the regression model.
3. Generate scatter plots between the residual series and the predicted values. Does the pattern appear to be random?
4. Generate a histogram of the residuals. Does it seem to be normally distributed? Perform a Q-Q test to double check the results.
5. Perform a RESET test to check if any higher order terms or interaction terms of the independent variables are missing from the regression model.
6. Request VIF statistics to check the level of multicollinearity. Is there any concern?
7. Perform a White test to check the homoskedasticity assumption. Request the Robust Standard Errors if the model does not pass the test.